

# Star and mesh networks

Revise 360 lesson plan — 60 minutes

|                 |                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------|
| Unit            | OCR J277 1.3 Computer networks, connections and protocols — Paper 1, Computer systems                             |
| Lesson          | Star and mesh networks (lesson 6 of the unit)                                                                     |
| Lesson ID       | nw-106                                                                                                            |
| Experience ID   | nw-106                                                                                                            |
| Plan ID         | PLAN-NW-L06                                                                                                       |
| Content version | 03a876d7fab7                                                                                                      |
| Website         | <a href="https://revise360.co.uk/experience.html?id=nw-106">https://revise360.co.uk/experience.html?id=nw-106</a> |

## Big question, outcomes and success criteria

**Big question** — Which shape of network would you fit here, and what are you giving up by choosing it?

### Prerequisites

The lesson before this one is nw-105. Pupils should already have finished it: the starter below retrieves from it.

Earlier in the unit: nw-101, nw-102, nw-103, nw-104.

### Assessable outcomes

1. **nw-106-o1** — know what a star network is
2. **nw-106-o2** — know what a mesh network is
3. **nw-106-o3** — understand that the internet is an example of a partial mesh network
4. **nw-106-o4** — know the advantages and disadvantages of star and mesh networks

## Vocabulary, resources and preparation

**Key vocabulary** — Topology, Star, Full mesh, Partial mesh

### Resources

|                   |                                                      |
|-------------------|------------------------------------------------------|
| SLIDES            | presentations/NW_L06_StarAndMeshNetworks_Lesson.pptx |
| WORKSHEET         | worksheets/NW_L06_StarAndMeshNetworks_Worksheet.docx |
| ONLINE EXPERIENCE | experience.html?id=nw-106                            |

### Before the lesson

1. Sign in on the **teacher dashboard** (teacher.html) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

### How pupils get in

1. Go to `topics.html` on `revise360.co.uk`.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.
3. Choose the topic, then this lesson's card.

**Worth knowing before you promise anything.** In this build `APP_CONFIG.backendUrl` is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

## Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

| MIN   | PHASE                                     | TEACHER                                                                                      | PUPILS                                                                          | EVIDENCE TO CHECK                                                                       |
|-------|-------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| 0-5   | <b>Retrieve</b>                           | Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites. | Answer the starter on the worksheet.                                            | The starter box on <code>NW_L06_StarAndMeshNetworks_Worksheet.docx</code> is filled in. |
| 5-8   | <b>Review the starter</b>                 | Take answers; correct the one most got wrong.                                                | Correct their own starter.                                                      | A correction written, not just a tick.                                                  |
| 8-18  | <b>Explain and model</b>                  | The new idea in two or three steps, with the example worked in front of the class.           | Listen and answer questions about the example.                                  | They can say why, not only what.                                                        |
| 18-21 | <b>Set up and settle</b>                  | Sign-in, launch, and what to write before anyone taps an answer.                             | Sign in, open the lesson, find Star topology.                                   | Everyone is in and on station 1.                                                        |
| 21-43 | <b>Online: work the stations in order</b> | Pupils read each wall, write the key fact and the challenge, then answer at the badge.       | Read each wall, write the key fact and the challenge, then answer at the badge. | The worksheet has writing on it for each of the 10 stations before the badge is tapped. |
| 43-48 | <b>Check understanding</b>                | The hinge question, hands down.                                                              | Answer the hinge question, hands down.                                          | The split across the options.                                                           |
| 48-56 | <b>Apply</b>                              | Exam practice on the worksheet, in silence.                                                  | Exam practice, in silence.                                                      | worksheet, Exam practice a) to d)                                                       |
| 56-60 | <b>Exit and set up the next lesson</b>    | The exit question and the retrieval link to next lesson.                                     | Write the exit answer.                                                          | One sentence that answers the big question.                                             |

## The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

### 1. Star topology

- Every device connects to a central switch with its own cable.
- All data travels through the switch, which sends it on to the device it is meant for.

**Challenge on this wall:** Why is it called a star network? Sketch one with five computers on your worksheet.

### 2. Advantages of star

- If one cable fails, only that device is affected.
- Easy to add a new device: just plug in one more cable.
- Fast and reliable: the switch only sends data where it needs to go.

**Challenge on this wall:** One cable in a star network is cut. Which devices lose their connection?

### 3. Disadvantages of star

- If the central switch fails, every device loses its connection. It is a single point of failure.
- Every device needs its own cable back to the switch, so it uses a lot of cable, which costs money.

**Challenge on this wall:** What is the biggest weakness of a star network? Explain why.

### 4. Joining star networks

- Bigger networks join several stars together by linking their switches with a cable.
- Devices on different switches communicate over this link.

**Challenge on this wall:** The link cable between two switches is cut. Who can still talk to whom?

### 6. Full mesh

- Every device is connected directly to every other device.
- Data can travel along many different routes to reach its destination.

**Challenge on this wall:** Count the cables in the picture. How many would you need to connect four devices in a full mesh?

### 7. Mesh: pros and cons

- No single point of failure: if a link fails, data is rerouted along another path.
- Can handle lots of traffic, because data is spread across many routes.
- Needs a lot of cable and is difficult to set up, so it is expensive.

**Challenge on this wall:** Why would a bank's data centre choose a mesh rather than a star?

### 8. Partial mesh: the internet

- In a partial mesh, devices connect to several others, but not to all of them.
- The internet is a partial mesh: routers link to a few other routers, so there is always another route.

**Challenge on this wall:** Why isn't the internet a full mesh?

### 9. Wireless mesh

- Wireless access points connect to each other by Wi-Fi. Each one must be in range of at least one other.
- Only one WAP needs a cable. Coverage is easy to extend, but data slows down as it hops between WAPs.

**Challenge on this wall:** Why might a wireless mesh be a good choice for a large house or an old building?

## Retrieval and the check for understanding

**Starter** — From last lesson, without notes: what is the internet, in one sentence? And what does a DNS server do when you type an address into a browser?

On the board. This starter is not printed on the pupils' worksheet: this lesson's sheet predates the current build and carries none.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

### Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (nw-106-s5-c4), so the wording a pupil sees matches the wording you say.

**Which single failure would stop every device from reaching the file server?**

**Correct:** Switch B fails

### What a pupil might choose instead

- Switch A fails

- The cable to Reception is cut
- The link cable between the switches is cut

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

**The course's own response to a wrong answer here:** The server is plugged into Switch B, so if Switch B fails nobody can reach it. Cutting the server's own cable would do the same.

**One response, not four.** one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

## Guided to independent practice

| STAGE     | ACTIVITIES | WHAT IT IS                                                                                                                |
|-----------|------------|---------------------------------------------------------------------------------------------------------------------------|
| Recognise | 20         | The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair.                   |
| Construct | 2          | Nothing is offered. The pupil classifies, orders, fills in a table or works a number out, and types or places the answer. |

## Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

## Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

## Delivering the online experience

### Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

### The stations, and what to record at each

| # | STATION            | ACTIVITIES | RECORD ON PAPER                                                                    |
|---|--------------------|------------|------------------------------------------------------------------------------------|
| 1 | Star topology      | 2          | Why is it called a star network? Sketch one with five computers on your worksheet. |
| 2 | Advantages of star | 2          | One cable in a star network is cut. Which devices lose their connection?           |

| #  | STATION                           | ACTIVITIES | RECORD ON PAPER                                                                                  |
|----|-----------------------------------|------------|--------------------------------------------------------------------------------------------------|
| 3  | <b>Disadvantages of star</b>      | 2          | What is the biggest weakness of a star network? Explain why.                                     |
| 4  | <b>Joining star networks</b>      | 2          | The link cable between two switches is cut. Who can still talk to whom?                          |
| 5  | <b>Break it!</b>                  | 4          | the answer to the station's challenge                                                            |
| 6  | <b>Full mesh</b>                  | 2          | Count the cables in the picture. How many would you need to connect four devices in a full mesh? |
| 7  | <b>Mesh: pros and cons</b>        | 2          | Why would a bank's data centre choose a mesh rather than a star?                                 |
| 8  | <b>Partial mesh: the internet</b> | 2          | Why isn't the internet a full mesh?                                                              |
| 9  | <b>Wireless mesh</b>              | 2          | Why might a wireless mesh be a good choice for a large house or an old building?                 |
| 10 | <b>Star vs mesh</b>               | 2          | the answer to the station's challenge                                                            |

### Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

### Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

### Assessment, evidence and unfinished work

| ID        | QUESTION                                                                                                                                         | MARKS | WHERE                                                  |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------------------------------------|
| nw-106-x1 | Explain the biggest weakness of a star network.                                                                                                  | 2     | worksheet, Exam practice a); deck, Exam practice slide |
| nw-106-x2 | Explain how a mesh network overcomes this weakness.                                                                                              | 2     | worksheet, Exam practice b); deck, Exam practice slide |
| nw-106-x3 | State one disadvantage of a full mesh network.                                                                                                   | 1     | worksheet, Exam practice c); deck, Exam practice slide |
| nw-106-x4 | A small business with 15 staff in one building is setting up a wired network. Should it use a star or a full mesh topology? Justify your answer. | 4     | worksheet, Exam practice d); deck, Exam practice slide |

### Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/nw-106.json`, which is not published with the website.

### Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **What is at the centre of a star network?** — A printer; A laptop
- **How does each device connect in a star network?** — With a cable to the next device along; With cables to every other device
- **One cable in a star network is cut. What happens?** — The whole network stops working; Every device slows down
- **Why does a star network perform well?** — Every device receives every message; Data travels around a ring
- **What is the biggest weakness of a star network?** — If one cable fails, every device loses its connection; It is hard to add new devices

### Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen

already shows what was and was not done, and that is the honest record.

## **Exit, next lesson and homework**

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### **Exit check**

One of these, written, not a show of hands:

- What a star network is
- What a mesh network is
- Why the internet is a partial mesh
- Advantages and disadvantages of star and mesh

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

### **Retrieval link to next lesson**

Next is nw-107, Revision HQ. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

### **Homework**

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

## **Shorter lessons, longer lessons and no internet**

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### **A 40 to 45 minute lesson**

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

### **If the internet or the devices are unavailable**

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

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